

Image quality evaluation

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Invalid Date

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1 Introduction

Human eyes can't detect if an image is different from another. But in some cases, it can be critical to know if the image is different from another, especially in compression domain where the goal is to conserve a good quality of image with a maximum compression. In this work, we will work on the image quality evaluation using several methods.

2 Methodology

To make image quality evaluation, we will use the MSE, PSNR, SSIM and LPIPS methods between two images, original image and modified image.

2.1 MSE (Mean Square Error)

This is a simple image quality evaluation that basically compute the mean between the two images and take the square of it. Mathematically, it can be represented by formula 1 for two images I and K of size $m \times n$.

$$\text{MSE} = \frac{1}{mn} \sum_{i=1}^m \sum_{j=1}^n [I(i, j) - K(i, j)]^2 \quad (1)$$

It give the sum of the square mean of each pixel, so a little value mean that the two images are very similar.

2.2 PSNR

The Peak Signal Noise Ratio (PSNR) is a measure that evaluate the quality of reconstructed images for instance after compression by comparing the maximal strength of the signal to the strength of the noise generated during reconstruction that affect the image.

Mathematically, it can be represented by formula 2 for two images I and K of size $m \times n$.

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right) \quad (2)$$

Where MAX_I^2 is the maximum value of a pixel.

As said in section 2.1, a little value of the MSE indicate that the two images are very similar. So as we have $\frac{\text{MAX}_I^2}{\text{MSE}}$, it means that when images are very similar, PSNR is high.

2.3 SSIM

The Structural Similarity Index (SSIM) is a perceptual metric used to quantify the similarity between two images, focusing on human vision. It measure the perceptual differences between two images, based on visible structure of images. So it's not a pixel-wise method, but more a method useful in term of visual perception of human.

Mathematically, it can be represented by formula 3 for two images I and K of size $m \times n$.

$$\text{SSIM}(I, K) = \frac{(2\mu_I\mu_K + C_1)(2\sigma_{IK} + C_2)}{(\mu_I^2 + \mu_K^2 + C_1)(\sigma_I^2 + \sigma_K^2 + C_2)} \quad (3)$$

With:

- μ_I the mean of image I
- μ_K the mean of image K
- σ_I^2 the variance of image I
- σ_K^2 the variance of image K
- σ_{IK} the covariance between image I and image K
- C_1, C_2 constant values to stabilize the division

If we suppose that image I is image K , the SSIM give us:

$$\text{SSIM}(I, I) = \frac{(2\mu_I\mu_I + C_1)(2\sigma_I^2 + C_2)}{(\mu_I^2 + \mu_I^2 + C_1)(\sigma_I^2 + \sigma_I^2 + C_2)} = \frac{(2\mu_I^2 + C_1)(2\sigma_I^2 + C_2)}{(2\mu_I^2 + C_1)(2\sigma_I^2 + C_2)} = 1$$

So a value closest to 1 indicate that the two images are perceptually very similar.

3 Implementation

For the implementation, I use `python` with [OpenCV](#) to load image and to make the jpeg compression.

Else, `numpy` library is used to make operations on image such as adding a gaussian noise, compute the MSE (ref section [2.1](#)) or the PSNR (ref section [2.2](#)).

In my code, I work only with images in range $[0, 255]$.

4 Results

To achieve my results, I have downloaded photos on internet and I have tried to not have digital image or generated image. On generated images or on digital images, perhaps the quality measures will be different due to another structure of our image, with no natural noise etc...

Original image



=====

Gaussian noise with sigma = 1

MSE: 1.33905029296875

PSNR: 46.86283472019738

SSIM: 0.9966978718780168

=====

Gaussian noise with sigma = 5

MSE: 24.74127197265625

PSNR: 34.19658337502638

SSIM: 0.9357708798077935

=====

Jpeg compression 90

MSE: 7.6191864013671875

PSNR: 39.31171762266033

SSIM: 0.9900354972307785

=====

Jpeg compression 50

MSE: 50.512420654296875

PSNR: 31.096821796132584

SSIM: 0.879297637296279

Original image



=====

Gaussian noise with sigma = 1

MSE: 1.3363761901855469

PSNR: 46.87151631527638

SSIM: 0.991116527849378

=====

Gaussian noise with sigma = 5

MSE: 24.782974243164062

PSNR: 34.189269353459615

SSIM: 0.8398421072431019

=====

Jpeg compression 90

MSE: 4.24200439453125

PSNR: 41.85509246777023

SSIM: 0.9729234131095675

=====

Jpeg compression 50

MSE: 12.729198455810547

PSNR: 37.08279303418355

SSIM: 0.9368902127255937

Original image



=====

Gaussian noise with sigma = 1

MSE: 1.302587

PSNR: 46.98273621327742

SSIM: 0.9992338967040004

=====

Gaussian noise with sigma = 5

MSE: 24.129637

PSNR: 34.30529572323515

SSIM: 0.9848276213038708

=====

Jpeg compression 90

MSE: 7.673377

PSNR: 39.28093824859601

SSIM: 0.9962652145352185

=====

Jpeg compression 50

MSE: 69.434526

PSNR: 29.715048857631558

SSIM: 0.9234532550599298

Original image



=====

Gaussian noise with sigma = 1

MSE: 1.331959

PSNR: 46.885895041626114

SSIM: 0.9878650671679702

=====

Gaussian noise with sigma = 5

MSE: 24.809624

PSNR: 34.184601784563455

SSIM: 0.7917796908133047

=====

Jpeg compression 90

MSE: 0.223223

PSNR: 54.64341420340431

SSIM: 0.9983469290136069

=====

Jpeg compression 50

MSE: 2.58476375

PSNR: 44.00659506575346

SSIM: 0.9904252758128195

5 Discussion

6 Conclusion